RAHUL MITRA

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EDUCATION

National University of Singapore **Aug 2023 – Jul 2027**
Bachelor of Engineering, Industrial and Systems Engineering (GPA: 4.37/5.00)

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.

WORK EXPERIENCE

Warehouse AI & Operations Research Engineer (Part-Time), STMicroelectronics (NUS Capstone) **Aug 2026 – Present**

- Architected a layout-agnostic AI put-away engine reading each site's shelf coordinates, dimensions, and operating rules, so one model fits any STMicroelectronics warehouse without per-site rework.

Global Operational AI Systems & Data Engineer (Part-Time), Abbott Laboratories (Contract) **Jun 2026 – Present**

- Developed ADAM, an offline tool-calling LLM assistant (Gemma 4 E4B on llama.cpp) for 6 spray dryers across 4 global sites, letting engineers self-serve answers that previously needed APC specialists.
- Built a Seeq signal-analytics toolkit (Kalman smoothing, spectral and lag analysis, change points, matrix profiles) and a 5-stage offline handbook RAG pipeline (OCR, ONNX embeddings, Chroma, LangChain).

Data Analytics & Product Management Intern, Abbott Laboratories (Converted to Contract) **Jan 2026 – Jun 2026**

- Engineered a SimPy discrete-event digital twin and OR-Tools CP-SAT solver for hybrid flow-shop scheduling, comparing 3 approaches (heuristic, MIP, genetic algorithm) to produce decision-ready schedules.
- Owned cloud deployment and production support for a team-built APC simulator used in manufacturing and engineer training, with Docker packaging, Azure App Service hosting, and documented handover.

Software Engineer (Full Stack & Cloud), People's Association (Pro bono) **Sep 2025 – Sep 2026**

- Designed, built, and deployed Churp, a resident-driven garden-plot balloting platform now live in Tengah and slated for national rollout, seeded by a S$20K Sparks Community Innovation Fund award.
- Owned most software development and deployment across admin and public Next.js frontends, a NestJS backend, and GIS mapping.
- Integrated Singpass Login and Myinfo (OIDC) to verify residents' identity and address, keeping balloting fair for 150 sign-ups at release.
- Provisioned 10 AWS services with Terraform (ECS Fargate, Multi-AZ RDS with PostGIS, WAFv2, Lambda, KMS), shipped via keyless GitHub Actions CI/CD across 5 repos with gated migrations and auto-rollback.

RESEARCH EXPERIENCE

Computer Vision Researcher, BlendED x Amazon Robotics (AI+X Global Challenge) **Jul 2026 – Oct 2026**

- Reproduced a published UNet super-resolution model in PyTorch from broken author code, training 12 models; matched reported results at 8x and came within 0.78 dB at 2x and 4x.
- Recognised by the project lead as one of the project's biggest contributors.

PROJECTS

Team Lead & Finalist, Swarmline: SG Defence Tech Hackathon (1,300+ applicants) **Sep 2026**

- Directed a 5-person team to architect decentralised drone-swarm coordination on correlated random-walk search, gradient-flow dispersion, delay-tolerant gossip, cross-inhibition consensus, and response-threshold task allocation.
- Implemented a deterministic Unreal Engine 5.8 C++ simulation with Claude Code; under a ground-link jam, 30 drones confirmed all 8 walking targets in 10 of 10 seeded runs, versus 2.1 of 8 on average for an operator-run fleet.

Finalist, Volt Pulse SG: SMU Hack For Cities (Top 8 Finalist of 50+ teams) **Jan 2026**

- Built an autonomous loop managing electricity and utility savings across multiple HDB households at once on SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

CO-CURRICULAR ACTIVITIES AND VOLUNTEER EXPERIENCES

Elderly Care Volunteer, NTUC Health Active Ageing Centre **Sep 2025 – Sep 2026**

- Coordinated weekly senior-engagement activities, including home visits to check on residents' wellbeing, hear life stories, and help with everyday tasks, building empathy for accessible, community-focused product design.

SKILLS AND CERTIFICATIONS

Technical: Solution architecture, AWS (ECS Fargate, RDS, ALB, WAFv2, Route 53, ACM, KMS, Secrets Manager), Terraform, AWS SAM, Azure, Docker, Bitbucket CI/CD, API design, Singpass Login & Myinfo (OIDC), RAG architecture, vector search, AI tools (Claude Code, ChatGPT, Codex, Gemini, ComfyUI), Supabase, GIS mapping, stakeholder discovery, manufacturing systems
Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery